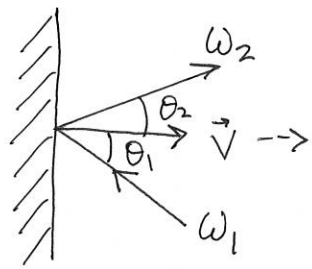
$\boxed{P1}$

K



In frame K: $k_1^i = \left(\frac{\hbar\omega_1}{c}, \hbar k_{1x}, \hbar k_{1y}, 0\right)$

$= \hbar\left(k_1, k_{1x}, k_{1y}, 0\right)$

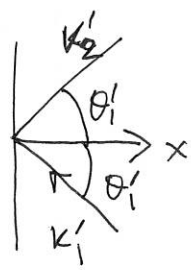
because $\omega_1 = c k_1 = c\sqrt{k_{1x}^2 + k_{1y}^2}$

Now, in the rest frame K' of the mirror

$$k_1'^i = \hbar\left(k_1', k_{1x}', k_{1y}', 0\right) \quad \text{and} \quad k_{1y}' = k_{1y}$$

and $\begin{pmatrix} \hbar k_1' \\ \hbar k_{1x}' \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} \hbar k_1 \\ \hbar k_{1x} \end{pmatrix}$ where $\beta = \frac{v}{c}$, $\gamma = \frac{1}{\sqrt{1-\beta^2}}$

K'



In K' $\theta_2' = \theta_1'$ and $k_{2x}' = -k_{1x}'$

$k_{2y}' = k_{1y}'$

and hence $k_2' = k_1'$

Hence after reflection we have

$$\begin{pmatrix} \hbar k_2' \\ \hbar k_{2x}' \end{pmatrix} = \begin{pmatrix} \hbar k_1' \\ -\hbar k_{1x}' \end{pmatrix}$$

To find k_2 in frame K (lab. frame) we transform back to K

$$\begin{pmatrix} \hbar k_2 \\ \hbar k_{2x} \end{pmatrix} = \gamma \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \begin{pmatrix} \hbar k_2' \\ \hbar k_{2x}' \end{pmatrix} = \gamma \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \begin{pmatrix} \hbar k_1' \\ -\hbar k_{1x}' \end{pmatrix}$$

But $\begin{pmatrix} k_1' \\ -k_{1x}' \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} k_1' \\ k_{1x}' \end{pmatrix}$

$$\Rightarrow \begin{pmatrix} k_2 \\ k_{2x} \end{pmatrix} = \gamma \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} k_1' \\ k_{1x}' \end{pmatrix} = \gamma \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} k_1 \\ k_{1x} \end{pmatrix}$$

$$= \gamma^2 \begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix} \begin{pmatrix} 1 & -\beta \\ \beta & -1 \end{pmatrix} \begin{pmatrix} k_1 \\ k_{1x} \end{pmatrix} = \gamma^2 \begin{pmatrix} 1+\beta^2 & -2\beta \\ 2\beta & -1-\beta^2 \end{pmatrix} \begin{pmatrix} k_1 \\ k_{1x} \end{pmatrix}$$

$$\begin{cases} k_2 = \gamma^2 \left((1+\beta^2) k_1 - 2\beta \, k_{1x} \right) & \omega_2 = c k_2 \\ k_{2x} = \gamma^2 \left(2\beta \, k_1 - (1+\beta^2) k_{1x} \right) & k_{2x} = k_2 \cos\theta_2 \end{cases}$$

But $k_{1x} = -k_1 \cos\theta_1 \Rightarrow \begin{cases} k_2 = \gamma^2 k_1 \left(1+\beta^2 + 2\beta \cos\theta_1\right) \\ k_{2x} = \gamma^2 k_1 \left(2\beta \cancel{\cos\theta_1} + (1+\beta^2)\cos\theta_1\right) \end{cases}$

(incident light moves in negative x-direction)

Thus $$\frac{k_{2x}}{k_2} = \cos\theta_2 = \cancel{\frac{1+\beta^2+2\beta\cos\theta_1}{(1+\beta^2)\cos\theta_1+2\beta}} \quad \frac{(1+\beta^2)\cos\theta_1 + 2\beta}{1+\beta^2+2\beta\cos\theta_1} \quad (1)$$

$$\omega_2 = \omega_1 \frac{1+\beta^2+2\beta\cos\theta_1}{1-\beta^2} = \left(1+\beta^2+2\beta\cos\theta_1\right) \frac{\omega_1}{1-\beta^2} \quad (2)$$

~~Observe from Eq. (1) that for $\beta \ll 1$,~~

$$\cancel{\cos\theta_2 \approx \frac{\cos\theta_1 + 2\beta}{1+2\beta\cos\theta_1} \approx 2(\cos\theta_1 + 2\beta)(1-2\beta\cos\theta_1) = \cos\theta_1 + 4\beta - 2\beta\cos^2\theta_1 = \cos\theta_1 + 2\beta\sin^2\theta_1}$$

Note that as $\beta \to 1 \; (V \to c)$

$$\cos\theta_2 \longrightarrow \frac{2(1+\cos\theta_1)}{2(1+\cos\theta_1)} = 1 \Rightarrow \theta_2 \to 0$$ angle of reflection approaches zero!

In the same limit $$\frac{\omega_2}{\omega_1} \Rightarrow \frac{2(1+\cos\theta_1)}{1-\beta} \to \infty$$

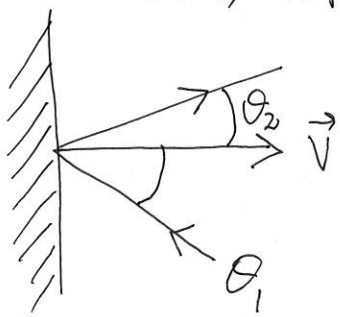
$$\cos\theta_2 - \cos\theta_1 = \frac{(1+\beta^2)\cos\theta_1 + 2\beta}{1+\beta^2+2\beta\cos\theta_1} - \cos\theta_1 = \frac{2\beta(1-\cos^2\theta_1)}{1+\beta^2+2\beta\cos\theta_1} =$$

$$= \frac{2\beta \sin^2\theta_1}{(1+\beta)^2 + 2\beta(\cos\theta_1 - 1)} > \frac{2\beta}{(1+\beta)^2}\sin^2\theta_1 > 0 \quad \text{for } \beta = \frac{v}{c} > 0$$

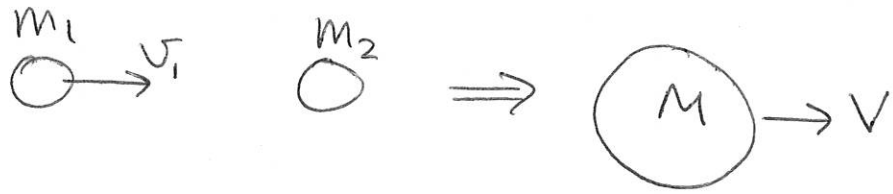
mirror moving along $+\hat{x}$ direction.

$\Rightarrow \cos\theta_2 > \cos\theta_1 \Rightarrow$ in the interval $\theta \in (0, \pi/2)$ $\theta_2 < \theta_1$.

Thus reflection looks like this:



P2



$$\left(\frac{E_1}{c}, \vec{p}_1\right) + \left(\frac{E_2}{c}, 0\right) = \left(\frac{E}{c}, \vec{p}\right)$$

$$\begin{cases} \dfrac{m_1 c^2}{\sqrt{1-\frac{v_1^2}{c^2}}} + m_2 c^2 = \dfrac{Mc^2}{\sqrt{1-\frac{V^2}{c^2}}} \\[2ex] \vec{p}_1 = \vec{p} \Rightarrow \dfrac{m_1 v_1}{\sqrt{1-\frac{v_1^2}{c^2}}} = \dfrac{MV}{\sqrt{1-\frac{V^2}{c^2}}} \end{cases}$$

$$\frac{M}{\sqrt{1-\frac{V^2}{c^2}}} = \frac{m_1}{\sqrt{1-\frac{v_1^2}{c^2}}} + m_2$$

$$\Rightarrow \frac{MV}{\sqrt{1-\frac{V^2}{c^2}}} = \frac{m_1 V}{\sqrt{1-\frac{v_1^2}{c^2}}} + m_2 V = \frac{m_1 v_1}{\sqrt{1-\frac{v_1^2}{c^2}}}$$

(This is center-of-mass velocity)

$$V = \frac{\dfrac{m_1 v_1}{\sqrt{1-\frac{v_1^2}{c^2}}}}{\dfrac{m_1}{\sqrt{1-\frac{v_1^2}{c^2}}} + m_2} = \frac{m_1 v_1}{m_1 + m_2\sqrt{1-\frac{v_1^2}{c^2}}}$$

$$\cancel{M = \frac{\sqrt{1-\frac{V^2}{c^2}}}{\sqrt{1-\frac{v_1^2}{c^2}}}\,(m_1+m_2) = \frac{\sqrt{1-\beta^2}}{\sqrt{1-\beta_1^2}}\,(m_1+m_2)}$$

$$M = \sqrt{\frac{1-\frac{v^2}{c^2}}{1-\frac{v_1^2}{c^2}}}\left(m_1 + m_2\sqrt{1-\frac{v_1^2}{c^2}}\right) =$$

$$= \sqrt{\frac{1-\beta^2}{1-\beta_1^2}}\left(m_1 + m_2\sqrt{1-\beta_1^2}\right)$$

$$\beta = \frac{v}{c} = \frac{m_1 \frac{v_1}{c}}{m_1 + m_2\sqrt{1-\frac{v_1^2}{c^2}}} = \frac{m_1 \beta_1}{m_1 + m_2\sqrt{1-\beta_1^2}}$$

$$\Rightarrow 1-\beta^2 = \frac{(m_1^2+m_2^2)(1-\beta_1^2) + 2m_1m_2\sqrt{1-\beta_1^2}}{\left(m_1 + m_2\sqrt{1-\beta_1^2}\right)^2}$$

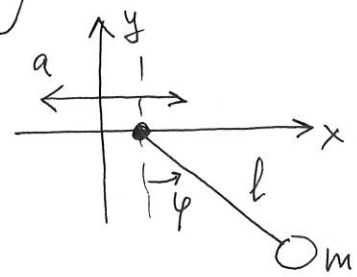
$$\Rightarrow M = \sqrt{\frac{m_1^2+m_2^2+\frac{2m_1m_2}{\sqrt{1-\beta_1^2}}}{\left(m_1+m_2\sqrt{1-\beta_1^2}\right)^2}}\left(m_1 + m_2\sqrt{1-\beta_1^2}\right)$$

$$= \sqrt{m_1^2+m_2^2+\frac{2m_1m_2}{\sqrt{1-\beta_1^2}}}$$

Check the non-relativistic limit $\beta_1 \to 0$:

$$M = \sqrt{m_1^2+m_2^2+2m_1m_2} = m_1+m_2 \quad \underline{\text{OK}}$$

P3



Kapitza pendulum with <u>horizontally</u> oscillating support

$X(t) = a\cos\gamma t$ is x-coordinate of the support point

Coordinates of m:

$$x = X(t) + l\sin\varphi$$

$$y = -l\cos\varphi$$

$$K = \frac{1}{2}m(\dot{x}^2+\dot{y}^2) = \frac{1}{2}m\left(\left(\dot{X} + l\cos\varphi\,\dot{\varphi}\right)^2 + \left(l\sin\varphi\,\dot{\varphi}\right)^2\right)$$

$$= \frac{1}{2}m\left(\dot{X}^2 + l^2\dot{\varphi}^2 + 2l\cos\varphi\,\dot{X}\dot{\varphi}\right)$$

φ-independent, can be dropped

Also $\dot{X}\cos\varphi\,\dot{\varphi} = \underbrace{\frac{d}{dt}\left(\dot{X}\sin\varphi\right)}_{\text{full derivative, can be dropped from the Lagrangian.}} - \ddot{X}\sin\varphi$

$$\Rightarrow K = \frac{1}{2}m\left(l^2\dot{\varphi} - 2l\ddot{X}\sin\varphi\right) = \frac{1}{2}m\left(l^2\dot{\varphi}^2 + 2la\gamma^2\cos(\gamma t)\sin\varphi\right)$$

$$U = +mgy = -mgl\cos\varphi \;;\; \ddot{X} = -a\gamma^2\cos(\gamma t)$$

$$\Rightarrow L = \frac{1}{2}ml^2\dot{\varphi}^2 + mla\gamma^2\cos(\gamma t)\sin\varphi + mgl\cos\varphi$$

EL equ: $\frac{d}{dt}\frac{\partial L}{\partial\dot{\varphi}} = \frac{\partial L}{\partial\varphi}$

$$ml^2\ddot{\varphi} = -mgl\sin\varphi - \underbrace{mla\gamma^2\cos(\gamma t)\cos\varphi}_{\text{oscillating force}}$$

$f(\varphi,t) = f_1(\varphi)\cos(\gamma t)$

Now we separate slow and fast motions: $\varphi = \underset{\text{slow}}{\phi} + \underset{\text{fast}}{\xi}$

Averaging over fast degree of freedom ξ we obtain effective potential $U_{eff}(\phi) = U(\phi) + \dfrac{f_1^2}{4ml^2\gamma^2} =$

$$= -mgl\cos\phi + \frac{(mla\gamma^2\cos\phi)^2}{4ml^2\gamma^2} =$$

$$= -mgl\cos\phi + \frac{1}{4}ma^2\gamma^2\cos^2\phi =$$

$$= mgl\left(-\cos\phi + \frac{a^2\gamma^2}{4gl}\cos^2\phi\right)$$

Stable equilibrium points: $\dfrac{dU_{eff}}{d\phi} = 0$

$$\frac{d^2U_{eff}}{d\phi^2} > 0.$$

$$\frac{d}{d\phi}\left(-\cos\phi + \frac{a^2\gamma^2}{4gl}\cos^2\phi\right) = \sin\phi - \frac{a^2\gamma^2}{2gl}\cos\phi\sin\phi = 0$$

$$\Rightarrow (a)\ \sin\phi = 0 \quad , (b)\ \frac{a^2\gamma^2}{2gl}\cos\phi = 1.$$

Solution (a): $\phi = 0$ or π.; (b) $\cos\phi = \dfrac{2gl}{a^2\gamma^2}$, possible if $2gl < a^2\gamma^2$.

$$\frac{d^2U_{eff}}{d\phi^2} = \cos\phi - \frac{a^2\gamma^2}{2gl}\left[(-\sin\phi)\sin\phi + \cos\phi\cdot\cos\phi\right] =$$

$$= \cos\phi - \frac{a^2\gamma^2}{2gl}\cos(2\phi)$$

Analyze $\frac{d^2 U_{eff}}{d\phi^2}$: (a) $\phi = 0 \Rightarrow \frac{d^2 U_{eff}}{d\phi^2} = 1 - \frac{a^2\gamma^2}{2gl} > 0$

for $a^2\gamma^2 < 2gl$.

$\phi = \pi \Rightarrow \frac{d^2 U_{eff}}{d\phi^2} = -1 - \frac{a^2\gamma^2}{2gl} < 0$

always $\Rightarrow$ <u>not possible</u>.

(b) $\cos\phi = \frac{2gl}{a^2\gamma^2} \Rightarrow \cos(2\phi) = 2\cos^2\phi - 1$

$$\Rightarrow \frac{d^2 U_{eff}}{d\phi^2} = \cos\phi - \frac{a^2\gamma^2}{2gl}\left(2\cos^2\phi - 1\right) =$$

$$= \frac{a^2\gamma^2}{2gl} + \frac{2gl}{a^2\gamma^2} - \frac{a^2\gamma^2}{gl}\left(\frac{2gl}{a^2\gamma^2}\right)^2 =$$

$$= \frac{a^2\gamma^2}{2gl} + \frac{2gl}{a^2\gamma^2} - 4\frac{gl}{a^2\gamma^2} =$$

$$= \frac{a^2\gamma^2}{2gl} - \frac{2gl}{a^2\gamma^2} > 0 \text{ for } \frac{a^2\gamma^2}{2gl} > \frac{2gl}{a^2\gamma^2}$$

$$\Rightarrow a^2\gamma^2 > 2gl.$$

Thus for $a^2\gamma^2 < 2gl$ the stable minimum is at $\phi = 0$.

for $a^2\gamma^2 > 2gl$ the stable minimum is at

$\pm\phi_0$, where $\phi_0 = \arccos\left(\frac{2gl}{a^2\gamma^2}\right)$.

Period of oscillations about non-trivial minimum:

Let $a^2y^2 > 2gl$ and consider $\phi = \phi_0 + \alpha$

$$\cos\phi_0 = \frac{2gl}{a^2y^2}.$$

$$U_{eff}(\phi) = U_{eff}(\phi_0+\alpha) \cong U_{eff}(\phi_0) + \frac{1}{2}U''_{eff}(\phi_0)\,\alpha^2$$

EL eqn: $$ml^2\ddot{\phi} = -\frac{d}{d\phi}U_{eff}(\phi)$$

$$ml^2\ddot{\alpha} = -\frac{d}{d\alpha}\left[\frac{1}{2}U''_{eff}(\phi_0)\,\alpha^2\right] = -U''_{eff}(\phi_0)\,\alpha.$$

$$\Rightarrow ml^2\ddot{\alpha} = -mgl\left(\frac{a^2y^2}{2gl} - \frac{2gl}{a^2y^2}\right)\alpha$$

$$\ddot{\alpha} = -\frac{g}{l}\left(\frac{a^2y^2}{2gl} - \frac{2gl}{a^2y^2}\right)\alpha = -\omega_0^2\,\alpha$$

$$\Rightarrow \omega_0 = \sqrt{\frac{g}{l}}\sqrt{\frac{a^2y^2}{2gl}\left[1-\left(\frac{2gl}{a^2y^2}\right)^2\right]}$$

$$\text{Period } T = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{l}{g}}\,\frac{1}{\sqrt{\frac{a^2y^2}{2gl} - \frac{2gl}{a^2y^2}}}$$

When $a^2y^2 \to 2gl$ from above, the period T diverges, while frequency ω_0 of oscillations about $\phi_0 = \arccos\left(\frac{2gl}{a^2y^2}\right)$ approaches zero.